

Problem #1

```
clear,clc
%{
Frank Sucansky, Nathan Moberg, Luis Duran AERE 1610, Lab 2 Problem 5.17
Purpose: rewrite a while loop into a for loop
%}
myProd = 1; %create a value for my prod
for i = [1:4]; %loop through range 1 -4
    num = input("Enter a number: "); %ask for user to input a number
    myProd = myProd * num; %multiply user number by myprod
end. %end loop
```

Output :

myProd = 1

Enter a number: 2

num = 2

myProd = 2

Enter a number: 6

num = 6

myProd = 12

Enter a number:

Comments:

- The original code will not display because there are semicolons at the end of every line.

Problem 5.21

The code in the while loop will be skipped under the condition that num is a negative number input by the user. Otherwise, the number will print "You have entered: " followed by the number rounded to the nearest tenth. In the event that a character other than a number is entered, the program will display an error.

Output :

N/A

Problem 4.22

```

clear,clc
%{
AERE 1610: Nathaniel Moberg, Frank Sucansky, Luis Duran: Lab2 Problem 4.22
purpose: write a menu to select either cylinder, circle, or rectangle, then
prompt the user for the relevant dimensions of the figure and calculate the
resulting area (or surface area)
%}
choice = input('Menu \n 1. Cylinder \n 2. Circle \n 3. Rectangle \n Please Choose One: '); %provides a display menu
with three options to choose from, prompts user to choose one
if choice==1 %if choice 1 is chosen
    radius = input('Radius of the cylinder: '); %prompts user for radius of cylinder
    height = input('Height of the cylinder: '); %prompts user for height of cylinder
    area = (2*pi*radius*height)+(2*pi*(radius.^2)); %area of a cylinder is 2*pi*r^2 (lateral) + 2*pi*r^2 (two faces)
    fprintf('The area is: %.2f', area) %print the resulting area
elseif choice==2 %if choice 2 is chosen
    radius = input('Radius of the circle: '); %prompts user for radius of circle
    area = pi*(radius.^2); %area of a circle is pi*r^2
    fprintf('The area is: %.2f', area) %print the resulting area
elseif choice==3 %if choice 3 is chosen
    length = input('Length of the rectangle: '); %prompts user for length of rectangle
    width = input('Width of the rectangle: '); %prompts user for width of rectangle
    area = length * width; %area of a rectangle is length*width
    fprintf('The area is: %.2f', area) %print the resulting area
else %else
    fprintf('Invalid choice, please choose 1, 2, or 3') %requests a valid input
end %ends block

```

Output :

```

1. Cylinder
2. Circle
3. Rectangle
Please Choose One: 1
Radius of the cylinder: 4
Height of the cylinder: 5
The area is: 226.19>>

```

Problem 5.17

```

clear,clc
%{
Frank Sucansky, Nathan Moberg, Luis Duran AERE 1610, Lab 2 Problem 5.17
Purpose: generate random numbers 0-50 and print those over greater or
equal to 25.

```

```
%}
num = 0; %create variable 0
count = 0; % Initialize count for the number of iterations
while num <= 25
    num = randi([0,50]); % Generate a random number between 0 and 50
    count = count + 1; % update count to keep track of how many tries it took
    fprintf("Number is: %d", num) %print number
End %end loop
fprintf("It took %d tries to get a number over 25", count) %print how many tries it took
```

Output :

Number is: 48

It took 1 tries to get a number over 25>>

Problem 5.3

```
clear, clc
%{
Frank Sucansky, Nathan Moberg, Luis Duran AERE 1610, Lab 2 Problem 5.3
Write a script that will prompt the user for a number in the inclusive range from 1 to 10
and error-check until the user enters a number in the correct range. Each time the user
enters an invalid number, it prints whether it is too low or too high. It then prints the valid
number.
%}
number = input('enter a number 1-10: '); %get initial number input
while number < 1 || number > 10 %loop through only checking whether a number is too
low or too high
    if number < 1 %check if number is too low
        fprintf('your number %g is too low', number) %print that number is too
        low
    elseif number > 10 %check if number is too high
        fprintf('your number %g is too high', number) %print that number is too
        high
    else %check if they input something invalid
        disp('your number is not valid') %display that it is invalid
    end %end the if statement
    number = input(' please enter a number 1-10: '); %ask for a number in range again
end %end the while statement
fprintf('your number is %g', number) %prints the valid number
```

Output:

```
enter a number 1-10: 0
your number 0 is too low please enter a number 1-10: 11
your number 11 is too high please enter a number 1-10: 5
```

Names of Present Group Members: [Type the document title]

AERE 161, Fall 2022

Lab #

your number is 5>>